2023 Winter Qual (Master)

III.

Let $G := VD^{-}U^{T}$

(i) $AGA = UDV^{T}VD^{-}U^{T}UDV$

$= UDD^{-}DV$

$= UDV = A$

(ii) $GAG = VD^{-}U^{T}UDV^{T}VD^{-}U^{T}$

$= VD^{-}DD^{-}U^{T}$

$= VD^{-}U^{T} = G$

(iii) $GA = VD^{-}U^{T}UDV^{T} = I$: symm

(iv) $AG = UDV^{T}VD^{-}U^{T} = I$: symm

Thus, G is a Moore-penrose g-inverse of A $\blacksquare$

$\boxed{2}$.

$$\mathrm{cov}(Ay, y^T By) = 2B\Sigma A\mu$$

$$= 2\sigma^2 \underbrace{BA}_{=0}\mu$$

$$= 0$$

$$Ay \perp y^T By \Rightarrow y^T Ay \perp y^T By$$

$\blacksquare$

$\boxed{3}$.

(a) (1) $= \frac{1}{2}\mu^T A \mu$

(2) $= A\Sigma$

$\boxtimes$

(b) $H_X = X(X^TX)^{-1}X^T$

$H_X^2 = H_X$: idempotent

$\text{rank}(H_X) = tr(H_X) = p$, $\frac{1}{2}\mu^T A \mu = \frac{1}{2}\beta^T X^T X \beta$

$y^T H_X y \sim \chi^2(p, \frac{1}{2}\beta^T X^T X \beta)$

$\boxtimes$

(c) $(I-H_X)^2 = I - H_X - H_X + H_X = I - H_X$

$tr(I-H_X) = n-p$

$\frac{1}{2}\mu^T A \mu = \frac{1}{2}\beta^T X^T (I-H_X) X \beta = 0$

$y^T(I-H_X)y \sim \chi^2(n-p)$

$\boxtimes$